

Tugas Praktikum 4

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Install RabbitMQ

```

$ docker run -d \
--hostname rabbitmq-host \
--name rabbitmq-container \
-p 5672:5672 \
-p 15672:15672 \
rabbitmq:3-management
Unable to find image 'rabbitmq:3-management' locally
3-management: Pulling from library/rabbitmq
20043866d3d5: Pull complete
3fad669ef11: Pull complete
7084ac8f4ab6: Pull complete
beb0d870d438: Pull complete
d63c4a88fdb7: Pull complete
efa05c660812: Pull complete
f4e7f62f4b80: Pull complete
5e1c03b67a78: Pull complete
14e6209d5fa8: Pull complete
d686b50ce172: Pull complete
Digest: sha256:56592c37766f3342496d4845efb5a1df782b5c63086ad017e2aae442311f69
Status: Downloaded newer image for rabbitmq:3-management
1194ef1f8e2787f500c132c32fbd25659e0e2390ef292befe82bd9e5312054e0b
$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED            STATUS              PORTS
1194ef1f8e278      rabbitmq:3-manage  "docker-entrypoint.s..." About a minute ago Up About a minute   4369/tcp, 5671/tcp, 0.0.0.0:5672->5672/tcp, 15671/tcp, 15671-156
92/tcp, 25672/tcp, 0.0.0.0:15672->15672/tcp, [::]:15672->15672/tcp rabbitmq-container

```

Akses RabbitMQ


RabbitMQ 3.13.7 Erlang 26.2.5.16

Refreshed 2026-05-16 21:01:05
Refresh every 5 seconds
Virtual host All
Cluster rabbit@rabbitmq-host
User guest Log out

Overview
Connections
Channels
Exchanges
Queues and Streams
Admin

Overview

Totals

Queued messages last minute 7
Currently idle
Message rates last minute 7
Currently idle
Global counts 7

Connections: 0
Channels: 0
Exchanges: 7
Queues: 0
Consumers: 0

Nodes

Name	File descriptors ?	Socket descriptors ?	Erlang processes	Memory ?	Disk space	Uptime	Cores	Info	Reset stats	+/-
rabbit@rabbitmq-host	44 (1048576 available)	0 (943629 available)	453 (1048576 available)	80 MiB (5.4 GiB high watermark)	146 GiB (48 MiB low watermark)	1h 44m	12	basic 2 79s	This node All nodes	

Churn statistics
Ports and contexts
Export definitions
Import definitions

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Code RabbitMQ.js

```
Help  Tugus-PWL
index.js rabbitmq.js x
book-services > config > as rabbitmq.js > RabbitMQService > connect
1  const amqp = require('amqplib');
2  class RabbitMQService {
3    constructor() {
4      this.connection = null;
5      this.channel = null;
6    }
7    Tabnine | Edit | Test | Explain | Document
8    async connect() {
9      try {
10        this.connection = await amqp.connect(process.env.RABBITMQ_URL || 'amqp://localhost');
11        this.channel = await this.connection.createChannel();
12        console.log('Connected to RabbitMQ');
13        return true;
14      } catch (error) {
15        console.error('Failed to connect to RabbitMQ:', error);
16        return false;
17      }
18    Tabnine | Edit | Test | Explain | Document
19    async sendToQueue(queueName, message) {
20      try {
21        if (!this.channel) {
22          await this.connect();
23        }
24        await this.channel.assertQueue(queueName, {
25          durable: true
26        });
27        this.channel.sendToQueue(
28          queueName,
29          Buffer.from(JSON.stringify(message)),
30          {
31            persistent: true
32          }
33        );
34        console.log('Message sent to queue ${queueName}:', message);
35        return true;
36      } catch (error) {
37        console.error('Error sending message to queue:', error);
38        return false;
39      }
40    Tabnine | Edit | Test | Explain | Document
41    async close() {
42      try {
43        if (this.channel) await this.channel.close();
44        if (this.connection) await this.connection.close();
45        console.log('RabbitMQ connection closed');
46      } catch (error) {
47        console.error('Error closing connection:', error);
48      }
49    }
50  }
51  module.exports = RabbitMQService;
52  }
53  }
54  }
55  }
56  }
57  }
58  }
59  }
60  }
61  }
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90  }
91  }
92  }
93  }
94  }
95  }
96  }
97  }
98  }
99  }
100 }
```

You have Maven installed on your system. Do you want to install the recommended 'Extension Pack for Java' extension from Microsoft for it?

Install Show Recommendations

Ln 12, Col 25 Spaces: 4 UTF-8 LF JavaScript Go Live Prettier

Cek & Test Message Broker

Get data - SRION WC's Workspace

Search

Overview Overview GET Get data GET Get data POST Get data + No environment

My Collection > Get data

POST http://localhost:3001/api/books Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "title": "Bismillah Bisa 2",
3   "author": "Febry D Fairuz",
4   "rating": 5,
5   "views": 2,
6   "is_free": false,
7   "language": "id-ID",
8   "sinopsis": "Pemrograman web adalah proses pengembangan aplikasi atau situs yang berjalan di browser, mengubah ide menjadi antarmuka interaktif menggunakan HTML, CSS, dan JavaScript",
9   "story": "Mencakup front-end dan back-end (PHP, Python, SQL), bidang ini berfokus pada logika, fungsionalitas, serta manajemen database untuk menciptakan website yang dinamis.",
10  "image": "https://www.ibik.ac.id/wp-content/uploads/2023/03/logo-ibikweb.png"
11 }
```

Body Cookies Headers 10 Test Results 0/1 201 Created 15 ms 1.02 KB Save Response

{ JSON Preview Visualize

```
1 {
2   "success": true,
3   "message": "Book created successfully",
4   "data": {
5     "id": 18,
6     "title": "Bismillah Bisa 2",
7     "author": "Febry D Fairuz",
8     "rating": 5,
9     "views": 2,
10    "is_free": false,
11    "language": "id-ID",
12    "sinopsis": "Pemrograman web adalah proses pengembangan aplikasi atau situs yang berjalan di browser, mengubah ide menjadi antarmuka interaktif menggunakan HTML, CSS, dan JavaScript",
13    "story": "Mencakup front-end dan back-end (PHP, Python, SQL), bidang ini berfokus pada logika, fungsionalitas, serta manajemen database untuk menciptakan website yang dinamis.",
14    "image": "https://www.ibik.ac.id/wp-content/uploads/2023/03/logo-ibikweb.png",
15    "updated_at": "2026-05-16T16:09:48.924Z",
16    "created_at": "2026-05-16T16:09:48.924Z"
17   }
18 }
```

Global Vault Tools

```

❖ injected env (8) from .env // tip: 🔑 secrets for agents [www.dotenvx.com]
❖ injected env (8) from .env // tip: 📁 multiple files { path: ['.env.local', '.env'] }
Executing (default): SELECT 1+1 AS result
✓ Koneksi ke database MySQL berhasil!
Connected to RabbitMQ
✓ RabbitMQ Connected
✓ Server running on port 3001
✓ API available at http://localhost:3001
Executing (default): INSERT INTO `books` (`id`,`title`,`author`,`rating`,`views`,`is_free`,`language`,`sinopsis`,`story`,`image`,`created_at`,`updated_at`) VALUES (DEFAULT,2,?,2,?,2,?,2,?,2,?,2,?)
🔥 Sending message to RabbitMQ...
✓ PRODUCER SENT MESSAGE
Message sent to queue book_created: {
  event: 'book.created',
  timestamp: '2026-05-16T16:07:13.673Z',
  data: {
    id: 17,
    title: 'Bismillah Bisa',
    author: 'Febry D Fairuz',
    is_free: false,
    language: 'id-ID'
  }
}
✓ RabbitMQ Result: true

```

Get messages

Warning: getting messages from a queue is a destructive action. ?

Ack Mode:

Encoding: ?

Messages:

Get Message(s)

Message 1

The server reported 0 messages remaining.

Exchange	(AMQP default)
Routing Key	book_created
Redelivered	0
Properties	delivery_mode: 2
	headers:
Payload	
160 bytes	
Encoding: string	{"event":"book.created","timestamp":"2026-05-16T16:07:13.673Z","data":{"id":17,"title":"Bismillah Bisa","author":"Febry D Fairuz","is_free":false,"language":"id-ID"}}

Learn more at <https://docs.docker.com/go/debug-cli/>

```

> docker exec -it rabbitmq-container rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
book_created  1

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → `docker debug rabbitmq-container`
 Learn more at <https://docs.docker.com/go/debug-cli/>

📁 ~/Documents/Semester 6/Tugas-PWL/user-services